Announcement: Submission of Lab Report 1

Course: Computer Network Sessional

Lab Title: Introduction to Network Hardware and Basic Networking Equipment

All students are hereby instructed to prepare and submit **Lab Report 1** based on the laboratory session on the introduction to network hardware and basic networking equipment.

The report should be written clearly and concisely, describing the observed equipment, their functions, and the practical activities performed during the lab session. Students must include relevant images of the observed items to support their descriptions.

Instructions for Preparing the Lab Report

Your report must contain the following sections in the specified order:

1. Cover Page

The cover page must include:

- Name of the university/institution
- Department name
- Course title: Computer Network Sessional
- Lab Report No.: 1
- Lab Title: Introduction to Network Hardware and Basic Networking Equipment
- Student Name
- Student ID
- Submission Date

2. Objective

Briefly state the objectives of the laboratory session. For example:

- To become familiar with basic computer networking equipment.
- To understand the functions of common network devices and accessories.
- To learn the construction and testing of a copper straight-through cable.
- To observe a server room and identify its basic components.

3. Equipment/Apparatus

List the equipment and tools observed or used during the lab session, including:

1. Switch
2. Router
3. Hub
4. Server
5. Personal Computer (PC)
6. Copper straight-through cable
7. RJ-45 connector
8. CAT cable cutter and crimping tool
9. Cable tester for checking the copper straight-through cable
10. Network Interface Card (NIC)
11. Network bridge
12. Server room and its relevant equipment

4. Lab Work and Observations

This is the **main part of the report**. Students must provide a short description of each observed item. For each device or tool, include:

- **Name:** The name of the device or equipment.
- **Description:** What it is and what it looks like.
- **Function:** Its primary purpose in a computer network.
- **Observation:** What you observed during the laboratory session, including any practical demonstration.
- **Image:** Add a relevant image of the observed equipment. Use your own photographs taken during the lab session whenever possible. If a photograph is unavailable, use a suitable reference image and provide its source.

For example, when describing a switch, explain that it connects multiple devices within a local area network (LAN) and forwards data to the appropriate destination using MAC addresses. Include an image of the observed switch.

For the practical activities, describe the following:

A. Copper Straight-Through Cable Construction

- Briefly explain the purpose of a straight-through cable.
- Describe the UTP/CAT cable, RJ-45 connector, and crimping tool used.
- Explain the steps followed to arrange the wires according to the applicable wiring standard (T568A or T568B), insert them into the connector, and crimp the connector.
- Include an image of the cable construction process or the completed cable, if available.

B. Copper Straight-Through Cable Testing

- Describe the cable tester used.
- Explain how the completed cable was connected to the tester.
- Record the observed indicator lights or test results.
- State whether the cable was functioning correctly according to the observed test results.
- Include an image of the cable tester or the testing process.

C. Server Room Inspection

- Briefly describe the server room and the equipment observed.
- Identify the visible networking and computing equipment, such as servers, switches, routers, racks, or cabling, where applicable.
- Explain the purpose of the observed equipment and the importance of a server room in network operation.
- Include relevant photographs of the server room or its equipment, if permitted.

Important: Write your observations in your own words based on what you actually saw and learned during the laboratory session. Do not simply copy textbook definitions. Keep each equipment description short, informative, and relevant.

5. Discussion

Write a short discussion summarizing what you learned from the laboratory session. Explain how the observed devices work together in a computer network and what you learned from constructing and testing a copper straight-through cable. You may also mention any difficulties encountered during the practical activities.

6. Conclusion

Conclude the report with a brief summary of the knowledge and practical skills gained from the session. Mention how the laboratory helped you understand basic network hardware, cable construction and testing, and server room infrastructure.

Formatting and Submission Requirements

- **Submission format:** PDF only.
- **File naming format:** Student_ID_Lab_Report_No_Student_Name.pdf
- **Example:** 1125871_Lab_report_1_Zahir_Rahman.pdf
- Ensure that all sections are arranged in the correct order.
- Use clear headings, readable fonts, proper spacing, and well-organized images with captions.
- Make sure all images are relevant to the equipment or activities being described.
- Check the report for spelling and grammatical errors before submission.
- Ensure that your name and student ID are correct on the cover page and in the file name.

Final Reminder

All students are requested to prepare the report according to the instructions above and submit it in PDF format. The report should demonstrate your understanding of the observed network hardware and the practical activities performed during the lab session.

Please follow the prescribed file naming format carefully to ensure proper identification of your submission.